

*MA.6.NSO.1.4***Benchmark****MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.**

Example: Michael has a lemonade stand which costs \$10 to start up. If he makes \$5 the first day, he can determine whether he made a profit so far by comparing $|-10|$ and $|5|$.

Benchmark Clarifications:

Clarification 1: Absolute value situations include distances, temperatures and finances.

Clarification 2: Problems involving calculations with absolute value are limited to two or fewer operations.

Clarification 3: Within this benchmark, the expectation is to use integers only.

Benchmark Clarifications:

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.3, MA.6.NSO.4.1, MA.6.NSO.4.2
- MA.6.GR.1.1, MA.6.GR.1.2, MA.6.GR.1.3
- MA.7.NSO.2.1

Terms from the K-12 Glossary

- Absolute Value
- Integers
- Number Line
- Rational Number
- Whole Number

Vertical Alignment**Previous Benchmarks**

- This is the first introduction to the concept of absolute value.

Next Benchmarks

- MA.8.NSO.1.5, MA.8.NSO.1.7, MA.912.DP.1.4

Purpose and Instructional Strategies

In previous courses, students plotted positive numbers on a number line and related addition of positive numbers to distance on a number line. In grade 6 accelerated, students determine and compare absolute values. Students will also use the concept of opposites when solving problems involving order of operations and absolute value. In future courses, students will use the order of operations with rational numbers, including exponents and radicals to solve multi-step mathematical and real-world problems.

- All values within this benchmark are limited to integers. This extends within the course to MA.7.NSO.2.1 where students will be expected to solve mathematical problems using the order of operations with rational numbers including absolute values.
- Instruction includes making connections in absolute value problems to direction, distance, or speed. This benchmark connects to finding the distance between two points on a coordinate plane with the same x - or y -coordinate.

- Instruction within absolute value contexts are not limited to distances, temperature and finances. Other situations could arise from a predetermined amount, or zero point, and then measuring above or below that amount (*MTR.7.1*).
 - For example, Leah eats on average 1200 calories in a day. On Wednesday, her caloric intake was 400 calories different than her average. What are her possible caloric intakes on Wednesday?
- Students should progress from solving problems using a concrete number line to solving problems abstractly. Students should represent equations with a visual model to illustrate their thinking. This will allow students to solidify the abstract concept through a pictorial representation. When students understand both methods and how they connect, students are often able to think more flexibly and reason through challenging problems successfully (*MTR.2.1, MTR.5.1*).
- Instruction includes the use of technology, including calculators.

Common Misconceptions or Errors

- Students may incorrectly state the absolute value of a negative number has a negative value. Instruction includes opportunities for students to talk about absolute value as distance in real-world scenarios (*MTR.7.1*).
 - For example, the odometer on my car reads 92,500 miles when I leave my house to drive 89 miles to Grandma's house. When I get to Grandma's house, the odometer reads 92,589 miles. When I turn around and drive home, which is the opposite direction, will my odometer count backward and read 92,500 again when I get home, or will it read 92,678 miles?
- Students may incorrectly assume distance is only referring to physically traveling between locations, such as walking, biking or driving. However, if we plot two values on a number line, this can also represent distance because we are determining how far away two points or values are from each other (*MTR.3.1*).

Strategies to Support Tiered Instruction

- Teacher provides instruction to reinforce the concept of absolute value being the distance of a number from zero.
 - For example, students plot integer values that represent temperatures on a number line and then record the number of units from zero.
- Teacher provides instruction for utilizing the absolute value symbols within the order of operations and refers to them as grouping symbols.
 - For example, when evaluating $-|6|$, first apply the absolute value of 6, then apply the factor of -1 to result in a solution of -6 , so that
$$\begin{aligned}-|6| &= (-1)(|6|) \\ &= (-1)(6) \\ &= -6.\end{aligned}$$
- Instruction for comparing absolute values of integers includes the use of pictorial representations or number lines to model the comparison and the use of key features of the model to discuss the problem, using contextual language when provided.

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.7.1)

On March 1, Mr. Lopez weighed 187 pounds. This was 12 pounds different than he weighed on January 1. On January 1, Mr. Lopez weighed 25 pounds different than the preceding October 1. What might Mr. Lopez have weighed on October 1? Explain why this question could have multiple answers.

Instructional Items

Instructional Item 1

The Philippine Trench is located 10,540 meters below sea level and the Tonga Trench is located 10,882 meters below sea level. Which trench has the greater depth and by how many meters?

Instructional Item 2

What is the value of the expression $7 - |-3|$?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered*